LUIS FERNANDO DA ROSA

AI/ML Engineer · Agentic Systems · LLM Pipelines · Production AI from 0→1

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WHAT I BRING

I build AI systems that ship, not prototypes that stall. Six years across agentic systems, LLM pipelines, and computer vision: designing multi-agent architectures, owning technical decisions end to end, and navigating the constraints that decide whether AI reaches production: cost, latency, security, and messy data. At **Golabs** I architected a multi-agent engineering system operating inside a private payments codebase, plus the evaluation harness that decides which model backs each agent. At **Arionkoder** I invented a novel HTML injection technique for LLM-based content extraction; at **Flow+** I led AI product delivery end to end across healthcare and productivity domains. I thrive where ownership beats process, iteration beats planning, and a working demo on Monday is worth more than a roadmap slide.

TECHNICAL TOOLKIT

Agents & LLMs: LangGraph, LangChain, OpenAI API, Azure OpenAI, Gemini API, Claude / Claude Code, RAG, LLM-as-Judge, Structured Outputs, Prompt Engineering

Agent Reliability: Multi-agent orchestration, request routing, eval harnesses (accuracy / consistency / cost), end-to-end tracing, model selection

ML / CV: TensorFlow, PyTorch, Keras, OpenCV, ONNX, Point Cloud (Open3D, Trimesh), OCR (Tesseract)

Data & Storage: Pandas, NumPy, Scikit-Learn, ChromaDB, Vector Search, Firebase

Infrastructure: Docker, Kubernetes (bare-metal), AWS Lambda, Azure, Python, TypeScript, Node.js, Linux, Git/GitHub automation

EXPERIENCE

AI Engineer · Golabs Tech

Jun 2026 – Present

- **Architected a multi-agent engineering system** inside a private payment-processing codebase, with specialist agents for planning, backend, frontend, API development, and documentation, plus a routing layer that classifies each request and dispatches it to the right agent.
- **Built the shared agent framework** that standardizes tooling, state handling, and execution contracts across the system, so adding a new agent no longer means rebuilding the same scaffolding.
- **Designed the evaluation harness** that scores every agent on correctness, run-to-run consistency, and cost per task, and used the results to select which model backs each agent.
- **Instrumented end-to-end tracing** across agent runs, so failures in non-deterministic pipelines can be located at the exact step that broke.
- **Deployed every agent on enterprise-hosted models**, keeping proprietary payments code off public APIs and meeting the security constraints of a regulated fintech environment.

AI Engineer · Arionkoder

Jun 2025 – May 2026

- **Invented a bot_id HTML injection technique** that lets LLMs identify semantic content regions (main content, modules, nav) without hand-crafted selectors, the core IP of the extraction product, applicable to 10+ structurally diverse municipal website templates.
- **Designed a 4-stage LLM pipeline** (region discovery → content extraction → structuring → LLM-as-judge validation) that automated CMS population end to end, cutting manual content entry from hours to minutes per page.
- **Reduced token consumption** by pre-filtering irrelevant HTML sections before LLM calls, directly lowering per-run cost and improving pipeline latency without sacrificing extraction accuracy.
- **Built an LLM-as-judge quality gate** that removed the need for human review of every output, compressing prompt iteration cycles from days to hours and enabling rapid accuracy improvements in production.

Senior AI Engineer · Flow+

Nov 2023 – May 2025

- **Led development of a developer burnout detection system** ingesting Bitbucket and Jira telemetry in real time, surfacing risk signals 2–3 weeks earlier than traditional HR indicators, the company's flagship AI product.
- **Led end-to-end AI product delivery** across projects in healthcare and productivity automation, from architecture through deployment, with full technical ownership and no dedicated engineering manager.

- **Contributed to company technical strategy and hiring**, defining engineering standards and evaluating AI stack decisions as part of core leadership in a small, fast-moving team.

Freelance AI Engineer · Independent
Oct 2024 – Jan 2025

- **Delivered a RAG pipeline** over financial and inventory document corpora using ChromaDB and LangChain, enabling analysts to query thousands of pages in seconds vs. manual review of hours.
- **Automated document classification, entity extraction, and summarization** for financial reports, eliminating a full manual processing step and improving data-entry accuracy for the client's stock management workflow.
- **Optimized prompt design and embedding strategy** to achieve reliable structured outputs from unstructured financial documents, reducing the post-processing correction rate.

Computer Vision Engineer · Pix Force
Feb 2023 – Dec 2023

- **Shipped a LiDAR point cloud processing system** (Open3D + Trimesh) for coal silo volume estimation with <5% error vs. manual surveying, replacing a dangerous manual process with fully automated AI measurement.
- **Built and deployed an OCR + GPT document analysis system** in Docker, reducing document processing time ~70% vs. manual review and enabling a client to process 3× the volume with the same headcount.
- **Developed and optimized deep learning models** in TensorFlow and PyTorch across 7+ production CV use cases, applying advanced data preprocessing to improve model accuracy and inference speed.

Computer Vision Engineer · GETTER
May 2022 – Jan 2023

- **Deployed AI vision models to IoT edge devices** with real-time automation at <200ms latency on resource-constrained hardware, enabling a production use case that was previously computationally impossible on-device.
- **Developed and optimized deep learning models** in TensorFlow and PyTorch across 3+ production CV use cases, leveraging quantization and pruning to enhance inference efficiency while maintaining high accuracy.
- **Designed Docker-based deployment pipelines** for scalable AI applications, standardizing the team's entire model delivery process and eliminating ad-hoc deployment failures.

Computer Vision Lead & Researcher · LAMIA Laboratory – UTFPR
Sep 2020 – Sep 2022

- **Led a 5-person CV research team** delivering 3 research projects, including Wood-Inspector, a wood quality inspection system for the Amazon basin that earned external recognition and patent documentation.
- **Managed a bare-metal Kubernetes cluster** supporting concurrent AI training workloads across multiple research projects, with zero managed infrastructure budget.
- **Designed, trained, and validated deep learning models** for image classification, segmentation, and pattern recognition, building the hands-on ML foundation that now underpins production system design.

Software Engineer · Silicon Village
Jul 2021 – Apr 2022

- **Built an event-driven serverless backend** (TypeScript, AWS Lambda, Firebase Functions) enabling the product to handle traffic spikes with zero infrastructure management overhead.
- **Applied Docker and Kubernetes** to automate deployments, eliminating manual release steps and improving deployment reliability across microservices.

EDUCATION

Technician in AI & Machine Learning — *Feb 2025 | UniCesumar, Medianeira, PR, Brazil*
B.Sc. Computer Science (Incomplete) — *2018 – 2023 | UTFPR, Santa Helena, PR, Brazil*

LANGUAGES

English — Full Professional · **Portuguese** — Native · **Spanish** — Elementary

NOTABLE PROJECT

Wood-Inspector Amazônia — Advanced CV system for wood quality inspection and classification deployed in the Amazon basin. Patent documented. clb.lamia.sh.utfpr.edu.br